

**FINAL REPORT
WORKSHOP: PYTHON.
TOPIC: "DEVELOPMENT OF A MACHINE LEARNING MODEL USING THE
PYTHON PROGRAMMING LANGUAGE"**

Prepared by:

Full name: Ramy Gamouh.

Status: University Student

Major: Statistics and Data Science

Supervisor: Ali mehizel

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OUTLINE



Contexte and Objectif

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CONCLUSION

CONTEXTE AND OBJECTIF

Offer: For the holiday season, the hypermarket is launching a Gold subscription priced at \$499 (instead of \$999), giving customers a 20% discount on all purchases. The offer is exclusive to existing customers and can be purchased either online or in-store.

Challenge

- Not all customers will subscribe
- Telemarketing can't call everyone (cost/time limits)

Goals

- Identify likely subscribers
- Estimate each customer's acceptance probability
- Understand key decision-driving factors
- Prioritize calls toward high-potential customers

Methodology

- Build a predictive model from customer history (purchases, frequency, behavior)
- Predict acceptance probability per customer
- Analyze key factors via feature importance

Commercial report

Target Variable

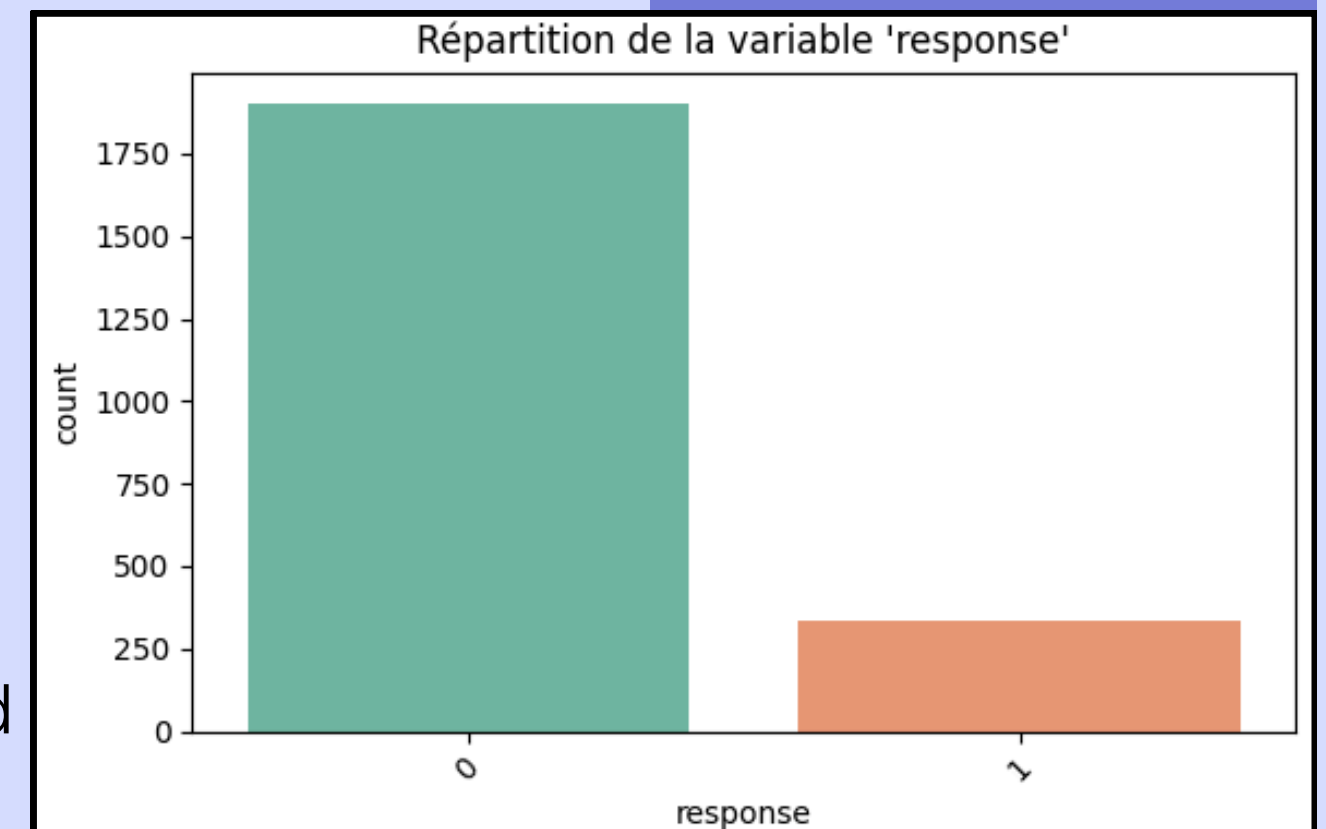
- Most customers did not accept the offer

Customer Profile

- Diverse in age and income; mostly no children at home
- Constant store activity (steady Recency)
- Spending across products is generally low
- Purchases mainly in-store/online, ~5–7 web visits/month
- Total spending: \$8–\$2,500, peaking at \$8–\$100

Education & Satisfaction

- Mostly degree-holders (Graduation, PhD, Master) and married
- Very few complaints → high satisfaction



Technical Report



Feature Engineering

- **child_home** = kids + teens at home
- **age** = derived from birth year, categorized
- **income_category**: income binning
- **education**: ordinal encoding

Results

- **Random Forest**: 64% precision
- **Stacking**: 67% precision (best)

Key Acceptance Factors (Random Forest Feature Importance)

- **Recency**: faster return → more engaged customer
- **NumCatalogPurchases**: more catalog purchases → more responsive
- **MntGoldProducts**: strong link to interest in premium products

Conclusion



Thanks to this predictive model, the telemarketing company can now focus its calls on customers with the highest probability scores, rather than contacting the entire customer base. This reduces costs and call time while maximizing the conversion rate. The recommended priority targeting focuses on loyal, educated customers with no complaints, high overall spending, and a particular interest in premium products — a segment that combines both a high probability of acceptance and strong profit potential for the hypermarket.



Thanks !

